sMDD (k = k*) vs MDD — DD Python, paired rows (same instance, DD type, width and reduction)
MDD: results_python_26_10_2026_16-30-00.csv | sMDD: results_python_semi_kopt_25_10_2026_19-35-30.csv
Columns are means over the paired rows; 'ratio' is the geometric mean of sMDD/MDD per row. Time = construction + reduction; Solve = longest/shortest path over the built diagram.
Bound means skip rows where either value is ±inf (empty restricted / unreachable terminal), counted in 'non-fin.'. 'sMDD +/=/-': rows where the sMDD bound is better / equal / worse.

Problem	DD	Red.	N	Time MDD	Time sMDD	ratio	Solve MDD	Solve sMDD	ratio	Nodes MDD	Nodes sMDD	ratio	Arcs MDD	Arcs sMDD	ratio	Bound MDD	Bound sMDD	non-fin.	sMDD +/=/-
Knapsack	Exact	no	24	31.21	48.40	1.56×	11.14	18.25	1.42×	3.26e+06	3.21e+06	0.58×	6.28e+06	2.25e+07	2.73×	13,215	13,215	0	0/24/0
	Exact	yes	21	9.49	17.63	1.85×	2.55	3.97	1.69×	872,545	872,545	1.00×	1.64e+06	6.17e+06	3.20×	7,599	7,599	0	0/21/0
	Restricted	no	147	9.22	12.35	1.38×	3.12	3.77	1.34×	1.21e+06	1.18e+06	0.59×	2.09e+06	4.97e+06	2.16×	11,275	11,275	0	0/147/0
	Restricted	yes	147	11.17	16.93	1.60×	3.01	3.46	1.46×	1.17e+06	1.17e+06	0.97×	2.05e+06	4.95e+06	2.50×	11,275	11,275	0	0/147/0
	RelaxPriority	no	147	10.53	11.13	1.25×	2.83	3.15	1.21×	1.21e+06	1.19e+06	0.60×	2.1e+06	4.11e+06	1.88×	300,155	299,432	0	46/101/0
	RelaxPriority	yes	147	12.66	14.71	1.39×	2.82	3.30	1.30×	1.17e+06	1.16e+06	0.96×	2.05e+06	4.05e+06	2.10×	300,155	299,432	0	46/101/0
Independent Set	Exact	no	14	6.86	94.08	1.80×	1.58	1.19	0.60×	872,766	777,701	0.64×	1.07e+06	1.31e+06	1.01×	27.93	27.93	0	0/14/0
	Exact	yes	14	7.91	95.51	1.66×	1.57	1.21	0.63×	872,766	777,701	0.64×	1.07e+06	1.31e+06	1.01×	27.93	27.93	0	0/14/0
	Restricted	no	113	1.29	15.19	2.39×	0.2774	0.3479	0.83×	174,315	151,362	0.79×	200,958	746,291	1.25×	24.16	23.35	0	0/93/20
	Restricted	yes	113	1.49	15.77	2.23×	0.2834	0.3610	0.84×	172,746	149,044	0.79×	198,885	738,811	1.24×	24.16	23.35	0	0/93/20
	RelaxPriority	no	114	1.86	17.61	2.02×	0.3266	0.4737	0.85×	229,372	221,356	0.84×	262,998	1.05e+06	1.23×	68.27	75.16	0	15/63/36
	RelaxPriority	yes	114	2.31	18.09	1.86×	0.3618	0.1732	0.52×	233,421	105,708	0.58×	267,904	229,335	0.75×	68.52	75.56	0	14/63/37
Set Cover	Exact	no	190	0.8765	71.09	0.77×	0.1521	11.57	1.46×	86,373	62,827	0.95×	166,820	2.47e+07	1.84×	3.63	3.63	0	0/190/0
	Exact	yes	188	0.8939	68.03	0.79×	0.0973	3.91	1.33×	54,687	50,218	0.99×	107,873	9.05e+06	1.70×	3.62	3.62	0	0/188/0
	Restricted	no	1199	0.4188	4.39	0.65×	0.0645	0.5205	1.19×	44,412	44,058	1.01×	86,484	1.57e+06	1.49×	3.72	3.72	0	0/1199/0
	Restricted	yes	1195	0.4637	5.01	0.69×	0.0547	0.2661	1.10×	37,482	39,216	1.01×	73,688	783,817	1.37×	3.72	3.72	0	0/1195/0
	RelaxPriority	no	1199	0.4403	4.53	0.66×	0.0653	0.5159	1.18×	44,539	45,332	1.01×	87,434	1.61e+06	1.50×	3.22	3.22	0	11/1176/12
	RelaxPriority	yes	1195	0.4872	5.17	0.70×	0.0571	0.2610	1.09×	37,878	39,471	1.01×	74,829	776,409	1.37×	3.22	3.22	0	8/1175/12
SOC Knapsack	Exact	no	84	4.16	44.88	2.86×	0.9679	7.99	2.13×	415,294	9,576	0.07×	706,467	1.73e+07	5.89×	791	791	0	0/84/0
	Exact	yes	84	5.18	69.78	3.57×	0.0215	5.76	25.51×	12,070	8,067	0.92×	23,577	1.28e+07	44.42×	791	791	0	0/84/0
	Restricted	no	530	0.2397	7.43	6.18×	0.0436	0.7748	4.43×	32,318	6,251	0.21×	39,681	2.4e+06	11.02×	688	771	0	351/179/0
	Restricted	yes	527	0.2666	8.58	7.41×	0.0011	0.3703	26.89×	1,579	4,203	1.83×	2,705	1.19e+06	45.35×	687	770	0	348/179/0
	RelaxPriority	no	531	1.17	7.61	3.24×	0.0550	0.6874	2.69×	35,509	8,069	0.22×	51,861	2.23e+06	5.68×	892	869	0	226/305/0
	RelaxPriority	yes	528	1.15	8.33	3.96×	0.0015	0.3413	10.28×	1,860	2,841	1.04×	3,379	1.08e+06	15.45×	891	868	0	224/304/0
Sequencing	Exact	no	5	22.70	1.83	0.08×	9.43	2.44	0.26×	524,274	65,536	0.13×	3.93e+06	524,288	0.13×	4,460	4,460	0	0/5/0
	Exact	yes	5	29.24	2.24	0.08×	3.56	2.22	0.62×	65,536	65,536	1.00×	524,288	524,288	1.00×	4,460	4,460	0	0/5/0
	Restricted	no	30	1.95	0.9678	0.69×	0.6578	1.06	1.95×	58,193	32,727	0.71×	322,737	235,429	0.91×	4,671	4,525	0	24/6/0
	Restricted	yes	30	2.25	1.13	0.70×	0.1729	1.03	6.58×	10,427	32,727	3.40×	58,206	235,429	4.41×	4,671	4,525	0	24/6/0
	RelaxPriority	no	30	6.07	3.28	0.85×	1.73	2.36	1.90×	76,052	45,190	0.77×	921,126	465,601	0.66×	2,695	2,056	0	5/0/25
	RelaxPriority	yes	30	6.75	3.59	0.85×	0.6315	2.31	5.39×	13,374	45,190	4.20×	161,845	465,601	3.63×	2,648	2,056	0	5/0/25